Nptel Online Certification Course
Indian Institute of Technology Kharagpur
Computer Vision
Assignment - Week 10

Number of questions: 10 **Total marks: 10x2=20**

QUESTION 1: **Type: MSQ**

Which of the following statements are true for an Artificial Neural Network?

a) A single neuron with two inputs and the signum activation function can implement an AND gate.

b) A single neuron with two inputs and the signum activation function can implement an XOR gate.

c) A single neuron with multiple inputs and the signum activation function acts as a linear discriminator in a multidimensional feature space for two linearly separable classes.

d) A single neuron with multiple inputs and the sigmoid activation function computes posterior of a class for a two-class classification problem in a multidimensional feature space.

Correct Answer: a, c, d
Detailed Solution:

a) True. Because an AND gate is a linearly separable function, a single neuron (perceptron) can easily find a decision boundary line that separates the output 1 from the output -1 (or 0).

b) False. Because an XOR gate is not linearly separable, meaning a single straight line cannot separate the true outputs from the false outputs in a 2D plane; it requires a multi-layer network.

c) True. Because the decision boundary of a single neuron with a signum activation function is defined by a linear combination of its inputs ($w^T x + b = 0$), which forms a hyperplane that splits the multidimensional space into two classes.

d) True. Because the sigmoid function mathematically models the posterior probability $P(\text{Class } 1 \mid x)$ under the assumption that the class-conditional distributions are normal with equal covariance matrices.

QUESTION 2: **Type: MCQ**

Consider the following table. The table shows the points on x-axis and its corresponding class.

Point	Class
0.8	1
1.2	1
1.9	1
2.5	2
3.0	1
3.5	1
4.0	2
4.8	2
5.5	3
6.2	3

We introduce two new points on x-axis, 2.7 and 4.2. Compute KNN (K=3) to determine their class.

a) 2.7 : Class 2 and 4.2: Class 1

b) 2.7 : Class 1 and 4.2: Class 2

c) 2.7 : Class 1 and 4.2: Class 3

d) 2.7 : Class 2 and 4.2: Class 2

Correct Answer: b)

Detailed Solution:

The point 2.7 is close to 2.5, 3.0 and 3.5. The maximum occurring class is 1 considering these points. Hence, the class assigned to 2.7 is 1. Similarly, for 4.2, the points are 3.5, 4.0 and 4.8.

QUESTION 3: **Type: MCQ**

Assume, you have 4 types of objects with 2 attributes each, as shown in the table, and you would like to form two clusters of the objects. Consider object A and object B as the initial centroids.

Table 1:

Object	Attribute 1	Attribute 2
A	1	1
B	2	1
C	4	3
D	5	4

What will be the elements in each clusters after the first iteration?

a) Cluster 1: A, Cluster 2: B, C, D

b) Cluster 1: A, D, Cluster 2: B, C

c) Cluster 1: A, B, Cluster 2: C, D

d) Cluster 1: A, C, Cluster 2: B, D

Correct Answer: a)

Detailed Solution:

Calculate the distance between each object with the cluster centroids.
Distance between Cluster 1 centroid (A) and B = 1
Distance between Cluster 1 centroid (A) and C = 3.61
Distance between Cluster 1 centroid (A) and D = 5
Distance between Cluster 2 centroid (B) and A = 1
Distance between Cluster 2 centroid (B) and C = 2.83
Distance between Cluster 2 centroid (B) and D = 4.24
Thus, the two clusters will be (A) and (B, C, D).

QUESTION 4: **Type: MCQ**

In the Naive Bayes algorithm, suppose that prior for class 1 is greater than class 2. How does the decision boundary shift from equally probable priors with respect to the region R1 (region for deciding class 1) or towards region R2 (region for deciding class 2)?

a) towards region R1.

b) towards region R2.

c) No shift in decision boundary.

d) It depends on the exact value of priors.

Correct Answer: b)

Detailed Solution:

As prior for class 1 is greater than class 2, we shift the decision boundary towards region R2 to preserve the prior probabilities proportion.

QUESTION 5: **Type: MCQ**

Assume, you want to cluster 7 observations into 3 clusters using K-Means clustering algorithm. After first iteration clusters, C1, C2, C3 has following observations:
C1: (2,2), (4,4), (6,6)
C2: (0,4), (4,0)
C3: (5,5), (9,9)
What will be the cluster centroids if you want to proceed for second iteration?

a) C1: (4,4), C2: (2,2), C3: (7,7)

b) C1: (6,6), C2: (4,4), C3: (9,9)

c) C1: (2,2), C2: (0,0), C3: (5,5)

d) C1: (2,2), C2: (0,4), C3: (5,5)

Correct Answer: a)
Detailed Solution:
Finding centroid for data points in cluster C1 = ((2+4+6)/3, (2+4+6)/3) = (4, 4)
Finding centroid for data points in cluster C2 = ((0+4)/2, (4+0)/2) = (2, 2)
Finding centroid for data points in cluster C3 = ((5+9)/2, (5+9)/2) = (7, 7)
Hence, C1: (4,4), C2: (2,2), C3: (7,7)

FOR QUESTIONS 6 AND 7:

Consider the following neural network. i1, i2 : Inputs; o1, o2 : Outputs. W1 = 0.15, W2 = 0.20, W3 = 0.25, W4 = 0.31, W5 = 0.40, W6 = 0.45, W7 = 0.50, W8 = 0.55.

If all the Activation Functions are ReLU, then answer the following question 6 and 7:.

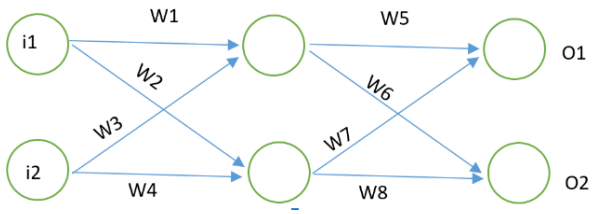


Figure 1: Feedforward neural network.

QUESTION 6: **Type: Comprehensive**

What will be the output for Input [1, 1]?

a) o1 : 0.21, o2 : 0.68

b) o1 : 0, o2 : 0

c) o1 : 0.37, o2 : 0.53

d) o1 : 0.42, o2 : 0.46

Correct Answer: d)

Detailed Solution:

i1 = 1, i2 = 1

h1 = ReLU(1*0.15 + 1*0.25) = ReLU(0.4) = 0.4

h2 = ReLU(1*0.20 + 1*0.31) = ReLU(0.51) = 0.51

o1 = ReLU(0.40*0.40 + 0.51*0.50) = ReLU(0.415) = 0.42

o2 = ReLU(0.40*0.45 + 0.51*0.55) = ReLU(0.46) = 0.46

QUESTION 7: **Type: Comprehensive**

What is the total error if o1 and o2 are 0 and 1, respectively? The error is given by $E = \frac{1}{2}(target - output)^2$.

a) 0.234

b) 0.468

c) 0.274

d) 0.548

Correct Answer: a)

Detailed Solution:

Error will be calculated as:

$E = \frac{1}{2}(0 - 0.42)^2 + \frac{1}{2}(1 - 0.46)^2 = 0.234$.

QUESTION 8: **Type: MCQ**

An optimization algorithm is updating a 197-dimensional parametric space to train a single-hidden-layer feedforward neural network. If the input layer accepts a 10-dimensional feature vector and the output layer yields a 2-dimensional vector, determine the number of neurons present in the hidden layer.

a) 12

b) 15

c) 20

d) 25

Correct Answer: b) 15

Detailed Solution:

Let h be the number of neurons in the hidden layer. The total number of parameters (weights and biases) in a network with a single hidden layer can be mathematically expressed by the formula: Total Parameters = $(d_{in} + 1) \cdot h + (h + 1) \cdot d_{out}$

$$
\begin{aligned}
197 &= (10 + 1) \cdot h + (h + 1) \cdot 2 \\
197 &= 11h + 2h + 2 \\
197 &= 13h + 2 \\
195 &= 13h \\
h &= \frac{195}{13} = 15
\end{aligned}
$$

Thus, the hidden layer contains exactly 15 neurons.

QUESTION 9: **Type: MSQ**

We want to classify whether a special wire is good or not. Consider the following table consisting of four samples.

Table 2:

Durability	Strength	Label
7	7	bad
7	4	bad
3	4	good
1	4	good
3	10	good
0	7	good
8	8	bad
6	3	bad

We introduce a new wire with durability 3 and strength 7. Which of the following statements are true after computing KNN (K=3)?

a) (3, 7) is a bad wire.

b) The nearest sample to (3, 7) is at a distance of 4.

c) (3, 7) is a good wire.

d) The nearest sample to (3, 7) is at a distance of 3.

Correct Answer: c), d)

Detailed Solution:

Calculate the distance between each sample and the new sample (3, 7).
Distance between (7, 7) and (3, 7) = 4
Distance between (7, 4) and (3, 7) = 5
Distance between (3, 4) and (3, 7) = 3
Distance between (1, 4) and (3, 7) = 3.6
Distance between (3, 10) and (3, 7) = 3
Distance between (0, 7) and (3, 7) = 3
Distance between (8, 8) and (3, 7) = 5.1
Distance between (6, 3) and (3, 7) = 5
It is closest to (3, 4), (3, 10) and (0, 7). Thus, the minimum distance is 3, and the sample gets classified as good.

QUESTION 10: **Type:MCQ**

In a computer vision classification task, a prototype-based model defines the representation of three classes (*Cat*, *Dog*, and *Fish*) using the following 5-dimensional feature vectors:

$$\mathbf{v}_{\text{Cat}} = [4, 1, 3, 0, 2]$$
$$\mathbf{v}_{\text{Dog}} = [5, 2, 4, 1, 1]$$
$$\mathbf{v}_{\text{Fish}} = [1, 0, 1, 5, 4]$$

Given an unknown input query feature vector $\mathbf{x} = [2, 1, 2, 4, 3]$, determine which class the input vector $\mathbf{x}$ belongs to based on the highest cosine similarity, and find its exact similarity value.

a) Fish, with a cosine similarity of 0.8412

b) Dog, with a cosine similarity of 0.7106

c) Fish, with a cosine similarity of 0.9231

d) Cat, with a cosine similarity of 0.7854

Correct Answer: c)

Detailed Solution:

Cosine similarity between vectors a and b is $\frac{a.b}{||a||\times||b||}$.
